

$k/n = 1/3$ exactly — H_1 dies here



$k = 1$ S^1 $k/n = 1/24$	$k = 2$ S^1 $k/n = 1/12$	$k = 3$ S^1 $k/n = 1/8$	$k = 4$ S^1 $k/n = 1/6$	$k = 5$ S^1 $k/n = 5/24$	$k = 6$ S^1 $k/n = 1/4$	$k = 7$ S^1 $k/n = 7/24$	$k = 8$ $V^7 S^2$ $k/n = 1/3$	$k = 9$ S^3 $k/n = 3/8$	$k = 10$ S^5 $k/n = 5/12$	$k = 11$ S^{11} $k/n = 11/24$	$k = 12$ full simplex $k/n = 1/2$
$H_1 = \mathbb{R}$	$H_1 = \mathbb{R}$	$H_1 = \mathbb{R}$	$H_1 = \mathbb{R}$	$H_1 = \mathbb{R}$	$H_1 = \mathbb{R}$	$H_1 = \mathbb{R}$	$H_1 = 0$	$H_1 = 0$	$H_1 = 0$	$H_1 = 0$	$H_1 = 0$

one loop, for every k below the boundary

higher spheres — invisible at $\text{maxdim} = 1$